

Student number: _____

2021 Preliminary Higher School Certificate Examination

Mathematics Standard

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen.
- NESA-approved calculators may be used
- A reference sheet is provided at the back of this paper

For questions in Section II show relevant mathematical reasoning and/or calculations

Total marks -100

Section I - 15 marks (pages 2 - 7)

- Attempt Questions 1 – 15
- Allow about 20 minutes for this section

Section II – 85 marks (pages 8 - 29)

- Attempt Questions 16 – 37
- Allow about 1 hour and 40 minutes for this section

Care has been taken to ensure that this paper is free of errors and that it follows the spirit of the syllabus. Some questions have been adapted from previous Preliminary HSC examinations and various sources, in an attempt to provide students with exposure to a broad range of questions.

Section 1

15 marks

Attempt Questions 1 – 15

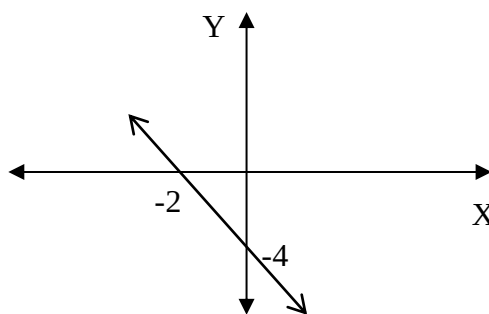
Allow about 20 minutes for this section.

Use the multiple – choice sheet for Questions 1 – 15

1. The algebraic expression $\frac{18m^6k^2}{36m^2k^8}$ in the simplest form is equal to:

(A) $2m^3k^4$ (B) $\frac{m^4}{2k^6}$ (C) $\frac{m^3}{2k^4}$ (D) $2m^4k^6$

2. The equation of the following linear function is:



- (A) $y = 2x - 2$ (B) $y = -2x - 4$ (C) $y = \frac{1}{2}x - 2$ (D) $y = -\frac{1}{2}x - 2$
3. An unbiased coin is tossed four times. On the second toss the result is tails. What is the probability the result of the fourth toss is tails?
- (A) $\frac{1}{8}$ (B) $\frac{1}{6}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2}$

4. Two litres of water has a mass of 2 kg. What would be the mass of 2 mL of water?

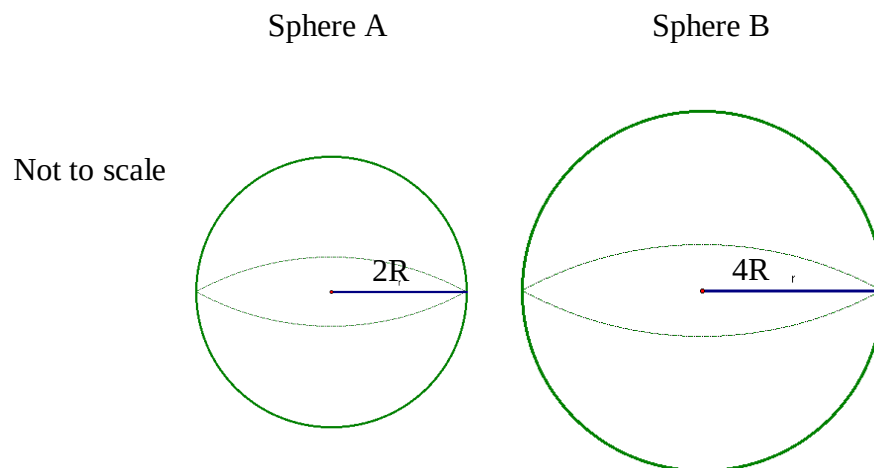
(A) 0.2 g

(B) 2 g

(C) 4 g

(D) 2000 g

5. The volume of sphere B is how many times larger than the volume of sphere A.



(A) 2

(B) 4

(C) 8

(D) 64

- 6 John and Beth sell products door-to-door. As a supervisor, John earns 18% commission on his sales, as well as 4.5% commission on Beth's sales. In a certain period, John sells goods for \$6800 and Beth sells \$3200 worth of goods. What is John's total commission?
- (A) \$1224
- (B) \$1241
- (C) \$1330
- (D) \$1368
7. 0.231×0.1999 to two significant figures is
- (A) 0.04
- (B) 0.046
- (C) 0.05
- (D) 0.0460
8. The co-ordinates of town A are latitude 31°S and longitude 122°E . Town B is due west of town A. Which of the following could be the latitude and longitude of town B?
- (A) 31°S , 110°E
- (B) 31°S , 148°E
- (C) 39°S , 122°E
- (D) 42°S , 122°E

9. The table below shows the domestic electricity prices for a particular town in NSW.

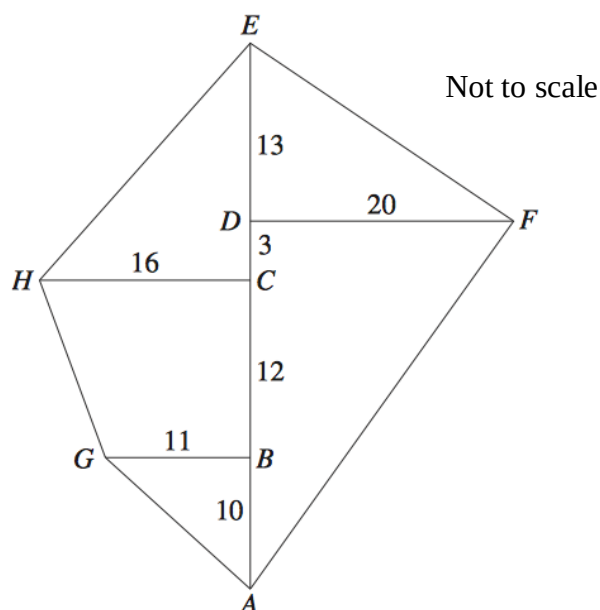
Electricity Tariffs	Rates in cents
Supply charge	85.16 c/day
Usage first 1685 kWh/quarter	31.42 c/kWh
Remaining usage/quarter	29.67 c/kWh

Calculate the amount charged, to the nearest dollar, for a quarter (91 days) if the usage is 2534 kWh.

- (A) \$309
- (B) \$467
- (C) \$652
- (D) \$859
10. Freda has an annuity that has a future value of \$600 000 on her retirement in 35 years. The annuity is invested at 9% p.a., with interest compounded annually. Calculate the present value of Freda's annuity.
- (A) \$760.64
- (B) \$980 000
- (C) \$22 678.98
- (D) \$29 391.64

11. A field diagram has been drawn from an offset survey of a major building site in Newport.

All units are in metres.



What is the total area of the above diagram, to the nearest metre?

- (A) 725 m^2
 (B) 768 m^2
 (C) 857 m^2
 (D) 925 m^2
12. Samuel is a 76kg male who had 5 standard drinks in 3 hours.

Calculate Samuel's BAC, correct to 2 decimal places, using the formula:

$BAC_{\text{Male}} = \frac{10N - 7.5H}{6.8M}$ where N is the number of standard drinks consumed, H is the number of hours of drinking, and M is the person's weight in kilograms.

- (A) 0.053 (B) 0.030 (C) 0.045 (D) 0.088
13. Solve the equation: $\frac{4p+2}{6} = 3$
- (A) $p = -2$ (B) $p = 4$ (C) $p = 2$ (D) $p = 3$

14. Given $T^2 = \frac{4p^2y}{k}$. Which of the following has p the subject of the formula.

(A) $p = \pm \frac{kT}{2\sqrt{y}}$ (B) $p = \pm \frac{T\sqrt{y}}{2\sqrt{k}}$ (C) $p = \pm \frac{T\sqrt{k}}{2\sqrt{y}}$ (D) $p = \pm \frac{T\sqrt{k}}{\sqrt{y}}$

15. Ned is preparing his **annual** budget for 2021. His expected income is:

- \$150 every week as a swimming coach
- A flat rate of 6% p.a. on an investment of \$6000

His planned expenses are:

- \$80 every week on transport
- \$18 every week on lunches
- \$130 every month on entertainment

He uses the spreadsheet below to determine how much money he can save in a year.

	A	B	C	D	E
1	Ned's Annual Budget for 2021				
2					
3	INCOME		EXPENSES		
4	Wages			Transport	
5	Interest			Lunches	
6				Entertainment	

Determine his savings for the year. Assume 52 weeks in a year.

- (A) \$250 (B) \$1 504 (C) \$2 560 (D) \$5 600

Student Number _____

Section II

85 marks

Attempt Questions 16-37

Allow about 1 hour and 40 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Your responses should include relevant mathematical reasoning and/or calculations.
Extra writing space is provided at the end of the paper. If you use this space, clearly indicate which question you are answering.

All necessary working should be shown in every question.

Question 16 (3 marks)

Medicare Levy
If your taxable income is less than or equal to \$18 600, your Medicare levy is nil. If your taxable income is: • more than \$18 600 but less than \$22 431, your Medicare levy is 25 cents for every dollar above \$18 600; • more than or equal to \$22 431, your Medicare levy is 2.35% of your taxable income.

- (a) Calculate the Medicare Levy, if Ben has a taxable income of \$21 890. **1**

- (b) If Mira has a taxable income of \$46 678. Calculate how much more Mira's Medicare levy will be compared to Ben's Medicare levy? **2**

Question 17 (2 marks)

In a certain test six students scored the following marks:

75 50 93 86 72 69

After further revision the test was repeated and the students' marks all increased by 8. When comparing the second set of results with the first, what effect will there be on the mean and the standard deviation?

Question 18 (4 marks)

TRAIN TIMETABLE

Station	pm	pm	pm	pm	pm
Parramatta	12:46	1:00	1:06	1:16	1:24
Harris Park	-	1:02	-	-	1:26
Granville	12:51	1:07	1:11	1:21	1:29
Cyde	-	-	1:17	-	1:35
Auburn	1:03	1:21	-	-	1:41
Redfern	-	1:35	-	1:47	1:54
Central	-	1:37	-	1:49	1:56
Town Hall	1:05	-	1:28	-	-
Wyndard	1:08	-	1:32	-	-

Belinda left home at 12:45 pm to catch a train at Parramatta, travelling to Auburn. Her office is a 20 minute walk from the Auburn railway station. It takes her 20 minutes to walk to Parramatta Station.

- (a) What is the earliest time she can arrive at her office?

2

- (b) Belinda drives to her office every Friday. The advertised fuel consumption of her car is 5.6 L/100km. The distance to her office from her home is 12 km. If fuel costs \$1.45 per litre, find the cost for her trip to the office.

2

Question 19 (5 marks)

Pitas and Joe work for a publishing company. Pitas is in charge of printing and is paid a salary of \$60 000. Joe is a salesman who sells books, each book sells for \$50 each. He is paid a monthly retainer of \$4 000 plus 10% of his total sales.

- (a) What is Pitas monthly pay? **1**

- (b) How many books must Joe sell to earn the same monthly pay as Pitas? **2**

- (c) Pitas is offered an alternative method of being paid, by piecework, where he would be paid \$1.20 per 1000 pages he prints. If the average book that they publish is 500 pages, how many books would he need to print to earn more than his annual current salary? **2**

Question 20 (4 marks)

A barrel has balls, identical in shape and size, which are numbered from 1 to 50. There are ten green, twelve blue, twelve red and sixteen yellow balls.

When one ball is drawn from the barrel, what is the probability that it is:

- (a) A number less than 36. **1**

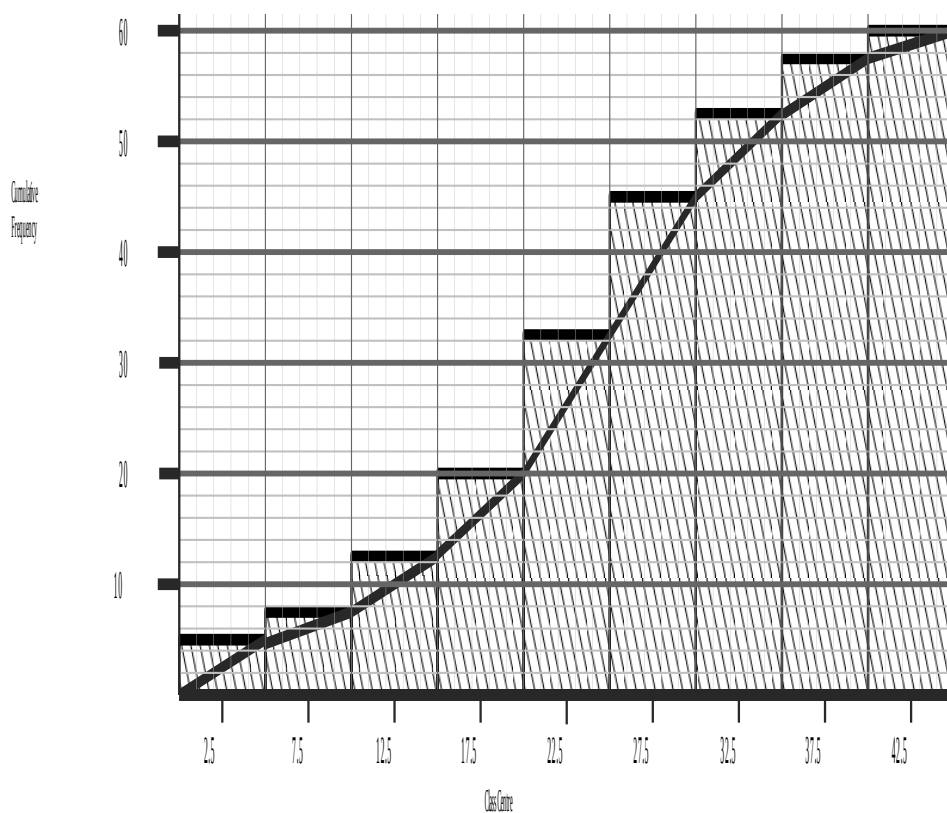
- (b) Not blue. **1**

- (c) Either blue or yellow. **1**

- (d) A number that is not divisible by seven. **1**

Question 21 (3 marks)

A cumulative frequency histogram and polygon are shown below.



(a) Estimate the median of the data.

1

(b) Estimate the interquartile range of the data.

2

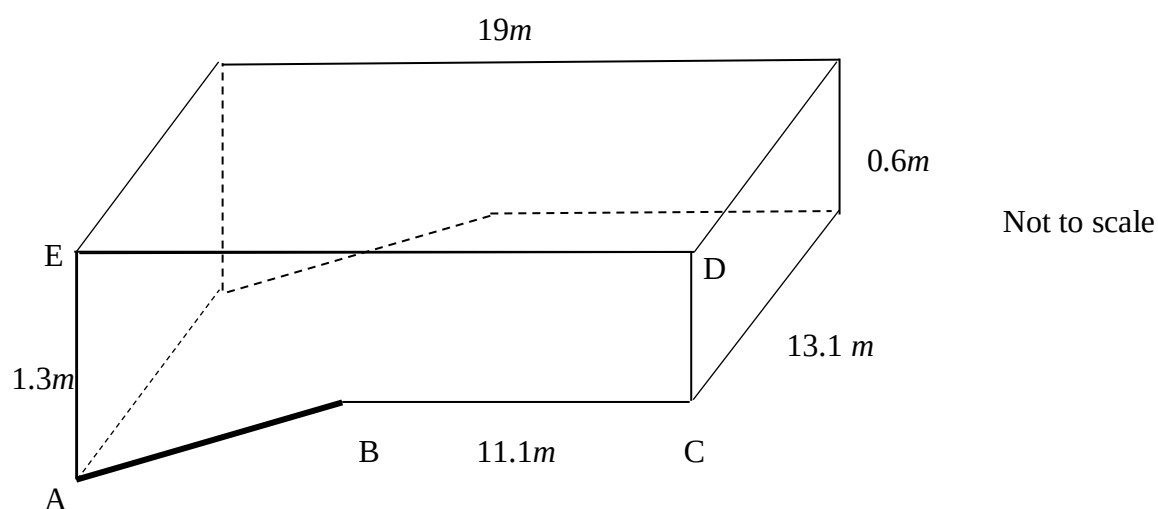
Question 22 (2 marks)

Use the table below to find the income tax payable on a gross salary of \$146 000, with allowable deductions of \$15 000.

<i>Taxable income</i>	<i>Tax on this income</i>
\$1 – \$6,000	Nil
\$6 001 – \$35 000	15c for each \$1 over \$6,000
\$35 001 – \$80 000	\$4 350 plus 30c for each \$1 over \$35,000
\$80 001 – \$180 000	\$17 850 plus 38c for each \$1 over \$80,000
\$180 001 and over	\$55 850 plus 45c for each \$1 over \$180,000

Question 23 (5 marks)

The diagram represents a water tank.



- (a) Find the length of AB to two decimal places. 1

- (b) Find the area of the shape ABCDE. 2

- (c) Find the number of litres of water to fill the water tank. ($1 \text{ m}^3 = 1 \text{ kL}$) 2

Question 24 (3 marks)

The table shows the amount of water used in a variety of situations.

Water usage	Volume	Time
Shower	170L	per 10 mins
Washing a car with a hose	450L	per car
Washing a car with a bucket	80L	per car
Washing clothes in a washing machine	140L	per load

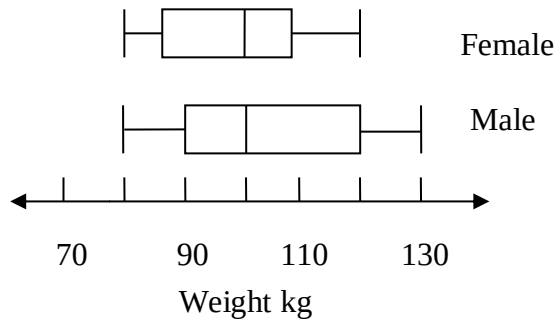
Water costs \$1.85/kilolitre.

Find the cost of water used if Mia showers for 5 minutes, does three loads of washing in a washing machine and washes her two cars with a hose.

Questions 16 – 24 are worth 31 marks

Question 25 (3 marks)

The box-and whisker plots shown are the weight of female and male rugby league players.



- (a) Find the five – number summary for male players.

2

- (b) What percentage of female players have a weight between 80 kg and 100 kg.

1

Question 26 (5 marks)

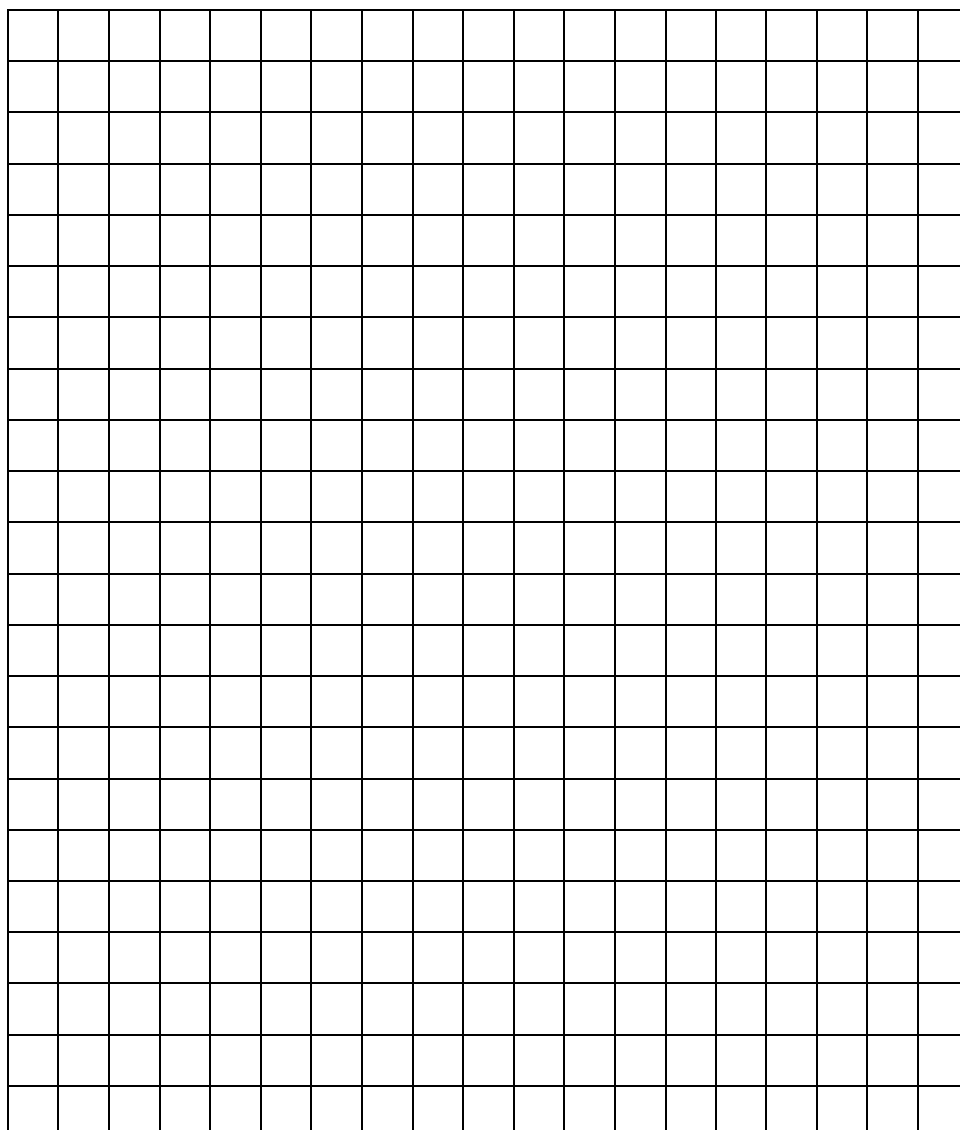
An infectious diseases physician used the formula $n = 84 - 6d$, where d is the number of days in self isolation to estimate the number of infections (n) in a local community.

- (a) Complete the following table for the function $n = 84 - 6d$ **1**

d	0	2	4	6	8	10	12	14
n								

- (b) Draw a graph of the function: $n = 84 - 6d$ **2**

Let the horizontal axis be the number of days



- (c) Find the gradient of the above function and explain what it represents? 2

Question 27 (4 marks)

There are 6 blue cards and 8 green cards in a box. Each card is identical in size and shape.

Two cards are selected from the box the first not being replaced in the box before the second is chosen.

- (a) Draw a tree diagram showing this information. 2

- (b) What is the probability that both cards selected are **NOT** blue. 1

- (c) What is the probability at least one of the cards selected is green. 1

Question 28 (4 marks)

This back to back stem-and-leaf plot shows the number of minutes a sample of people had to wait to get tested for a Covid-19 test on a Thursday and Friday.

Thursday	Stem	Friday
7 3 2	0	5 8 8
9 7 5 4 2	1	4 6 7 9
1 0 0	2	N 8 8 9
6 6 6 2	3	2 5
	4	3 4 5

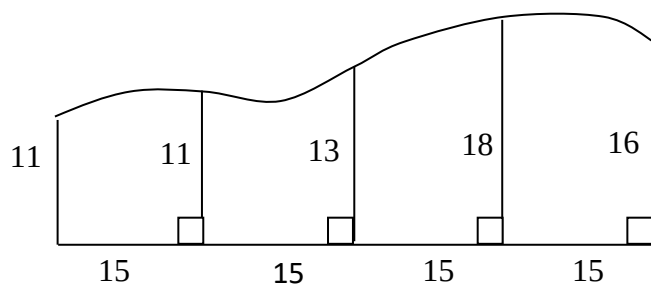
- (a) Determine the range, mode and median of the waiting times on Thursday. 2

- (b) One value is missing for Friday, indicated by the letter N. What is the value of N, if the mean waiting time for Friday is 24.625 minutes. 2

Question 29 (3 marks)

Mr Chen's backyard was measured as shown below. All measurements are in metres.

NOT TO SCALE



Using four applications of the trapezoidal rule, calculate the area of the backyard.

3

Question 30 (5 marks)

An electronic item costing \$3 800 depreciated by \$14 per year. Using the straight line method, determine:

(a) Its value after 5 years.

1

(b) When is the electronic item expected to reach a value of \$3 478?

Give your answer to the nearest month.

2

(c) Find the value of the GST (Goods and Services Tax) if 10% GST is included in the \$3 800.

2

Question 31 (4 marks)

Nick and Nicole ride a tandem bike. They can cycle a 35 km circuit with an average speed of 45 km per hour.

- (a) How many minutes did they take to cycle the 35 km? (Answer to the nearest minute) 2

- (b) Due to improved training techniques, they can now achieve an average speed of 58 km/h whilst completing the 35 km circuit.

How many minutes has their time improved to complete the circuit? 2

Question 32 (5 marks)

- (a) Susan's car uses 14 litres of petrol to travel 176 km. Petrol costs 146 cents per litre.

How far can she travel using \$66.60 worth of petrol (answer to the nearest km). 3

- (b) Jack drives the same car as Susan. When he was travelling at 70 km/h he saw an object on the road and applied his brakes stopping 18.6 m later. Find his reaction distance, given he has a reaction time of 1.8 seconds. Answer to the nearest metre? **1**

- (c) Find his stopping distance? **1**

Questions 16 – 32 are worth 64 marks

Question 33 (4 marks)

George works for two book stores. He works from 2:00 pm to 8:30 pm on Thursday evenings for 'Great Books'. He works from 8:30 am to 10:30 am on a Saturday and 9:00 am to 4:00 pm on Sundays for 'My Books'. The rates paid by each employer are listed below.

Great Books	
TIME	RATE
For the first 3 hours / day	\$13.00 / hour
For the next 3 hours / day	\$14.60 / hour
Any extra hours per day	\$15.60 / hour
Uniform allowance for all employees = \$4.50/ hour	

My Books
Casual rates
Weekdays \$13.50 / hour
Saturday - time and a half Sunday - double time

- (a) What are George's weekly earnings from 'Great Books'?

2

- (b) What are his total earnings for one week?

2

Question 34 (3 marks)

Compounded values of \$1

Period	Interest rate per period				
	1%	2%	3%	4%	5%
1	1.010	1.020	1.030	1.040	1.050
2	1.020	1.040	1.1061	1.082	1.103
3	1.030	1.061	1.093	1.125	1.158
4	1.041	1.082	1.126	1.170	1.216
5	1.051	1.104	1.159	1.217	1.276
6	1.062	1.126	1.194	1.194	1.340

- (a) Use the table to calculate the final value of \$1 590 invested at 5% p.a. for 4 years. **1**

- (b) Peta hopes to save \$6 700 in 2 years to buy a car. She finds an investment of 4% p.a. with interest compounded half-yearly. Using the table, calculate the amount of money she must invest to generate \$6 700 in this time. **2**

Question 35 (5 marks)

Betty and Garry want to purchase a new entertainment system. The price of the entertainment system is \$ 12 678. They decide on the following finance option:

- Deposit of \$560
- Interest on balance charged at a flat rate 13.5% p.a. for 4 years

They will repay the loan with equal monthly instalments over the 4 years.

- (a) Calculate the amount of interest charged on the loan. **2**

- (b) Calculate the monthly repayment they will need to make. **2**

- (c) Calculate the amount paid for the entertainment system. **1**

Question 36 (5 marks)

Cathy flew from Lord Howe Island to Juneau (Alaska). Lord Howe Island is 10.5 hours ahead of UTC (Coordinated Universal Time) and Juneau is 9 hours behind UTC. The plane stops at the Cook Islands (10 hours behind UTC) for a three hour stop over.

The flight from Lord Howe Island to the Cook Islands takes 5 hours and the flight from the Cook Islands to Juneau takes 8 hours.

- (a) What time is it in the Cook Islands when it is Wednesday 3 p.m. in Lord Howe Island?

2

- (b) If Cathy arrives in Juneau at 5:30 am Wednesday, Alaskan time, what time did the plane leave Lord Howe Island?

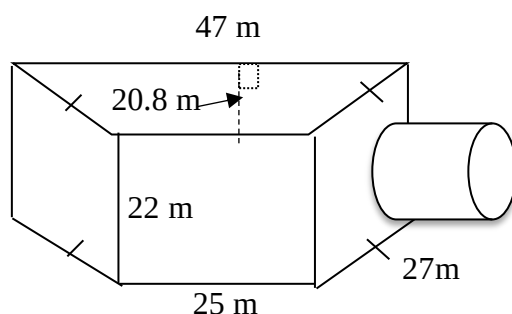
3

Question 37 (4 marks)

The following solid has a closed cylinder attached to a trapezoidal prism.

The cylinder is 29 m long and diameter of 20 m.

Not to scale



- (a) Find the surface area of the solid, including the bottom, correct to the nearest m^2 . **3**

- (b) If a 20-litre can of paint covers 170m^2 , find the number of 20-litre cans of paint needed to paint the outside of the solid with one coat. **1**

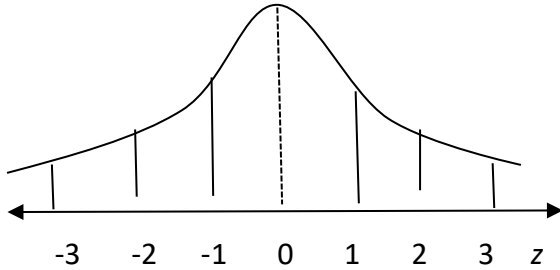
End of Exam

Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

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Mathematics Standard 1 and Standard 2 Reference sheet

Measurement Limits of accuracy Absolute error = $\frac{1}{2} \times \text{precision}$ Upper bound = measurement + absolute error Lower bound = measurement – absolute error Length $l = \frac{\theta}{360} \times 2\pi r$ Area $A = \frac{\theta}{360} \pi r^2$ $A = \frac{h}{2} (a + b)$ $A \approx \frac{h}{2} (d_f + d_l)$ Surface area $A = 2\pi r^2 + 2\pi r h$ $A = 4\pi r^2$ Volume $V = \frac{1}{3} Ah$ $V = \frac{4}{3} \pi r^3$ Trigonometry $\sin A = \text{opp/hyp}$, $\cos A = \text{adj/hyp}$, $\tan A = \text{opp/adj}$ $A = \frac{1}{2} ab \sin C$ $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ $c^2 = a^2 + b^2 - 2ab \cos C$ $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$	Financial Mathematics $FV = PV(1 + R)^n$ Straight-line method for depreciation $S = V_0 - Dn$ Declining-balance formula for depreciation $S = V_0(1 - r)^n$ Statistical Analysis An outlier is a score less than $Q_1 - 1.5 \times IQR$ or more than $Q_3 + 1.5 \times IQR$ $z = \frac{x - \bar{x}}{s}$ Normal distribution  - Approximately 68% of scores have z-scores between -1 and 1 - Approximately 95% of scores have z-scores between -2 and 2 - Approximately 99.7% of scores have z-scores between -3 and 3
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